

From Portfolio Breadth to Platform Evidence

competitive strategy, telemetry analytics, anomaly detection, RL benchmarks, decision evaluation, dashboard design and platform recommendations.

CAPSTONE THESIS

InGen's most defensible opportunity is not any single robot. It is a shared physical-intelligence and evaluation layer that turns a broad portfolio into compounding evidence.

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Executive agenda: from strategy to operating model

- 1

Strategic position
Competitive landscape, strategic gaps and the PIC 2.0 portfolio thesis.
- 2

Evidence stack
Telemetry analytics, anomaly detection, RL benchmarks and decision-eval reliability.
- 3

Portfolio application
Dashboard overview, platform-specific recommendations, limitations and next steps.

Single Most Important Conclusion

Portfolio breadth only becomes a strategic advantage when the products share a common evidence system: telemetry contracts, evaluation contracts, safety rules, human governance and deployment metrics.

Phase	Weeks	Evidence produced
Strategy	1–2	Physical-AI landscape, 22-vendor matrix, strategic gaps
Analytics	3–4	Rover telemetry, features, injected faults, anomaly benchmark
Learning	5	PPO/SAC + multi-agent benchmarks with caveats
Decision	6	50 scenarios, 150 seeded judgments, alpha/kappa
Integration	7–8	Dashboard, LSTM battery follow-up, capstone synthesis

Competitive landscape snapshot: breadth is rare, evidence is rarer

Vendors reviewed
22
humanoid ·
service ·
security

Market pattern
**fragmented
embodiment-led**

InGen opening
**shared layer
one intelligence
layer**

Strategic read
Most competitors center on one embodiment, workflow or vertical. InGen can win by making shared architecture visible in telemetry, evaluation, governance and deployment practice, not only in product positioning.

Competitive cluster	Observed pattern	Strategic gap	Evidence I would prioritize
Humanoids	Capital-intensive embodiment race	Hard to out-claim general autonomy	Bounded skills, sim-to-real gates, intervention rate
Care / education	Trust and outcomes matter more than novelty	Conversational ambiguity is under-evaluated	Escalation rules, consent, learning/care outcomes
Security / patrol	Operator burden dominates customer trust	False-alert cost often hidden	Alert budget, calibrated uncertainty, review time
InGen portfolio	Multiple products can share infrastructure	Breadth risks duplicated engineering debt	Reuse ratio: schemas, harnesses, safety, monitoring

My recommendation
Measure PIC 2.0 reuse explicitly in every release: inherited schemas, evaluation harnesses, uncertainty modules, safety rules and monitoring components.

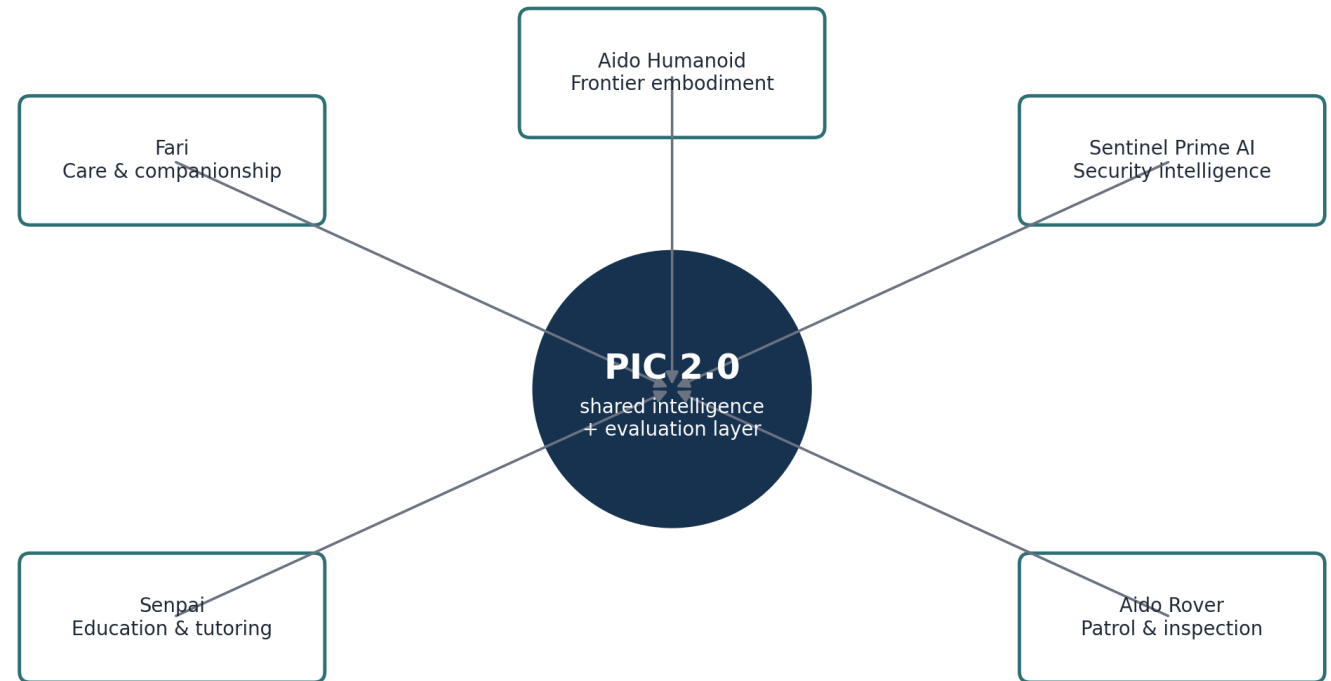
Portfolio compounding: PIC 2.0 as the evidence spine

What compounds

Telemetry schemas · evaluation contracts · safety rules · uncertainty fields · release evidence · dashboarded operational outcomes

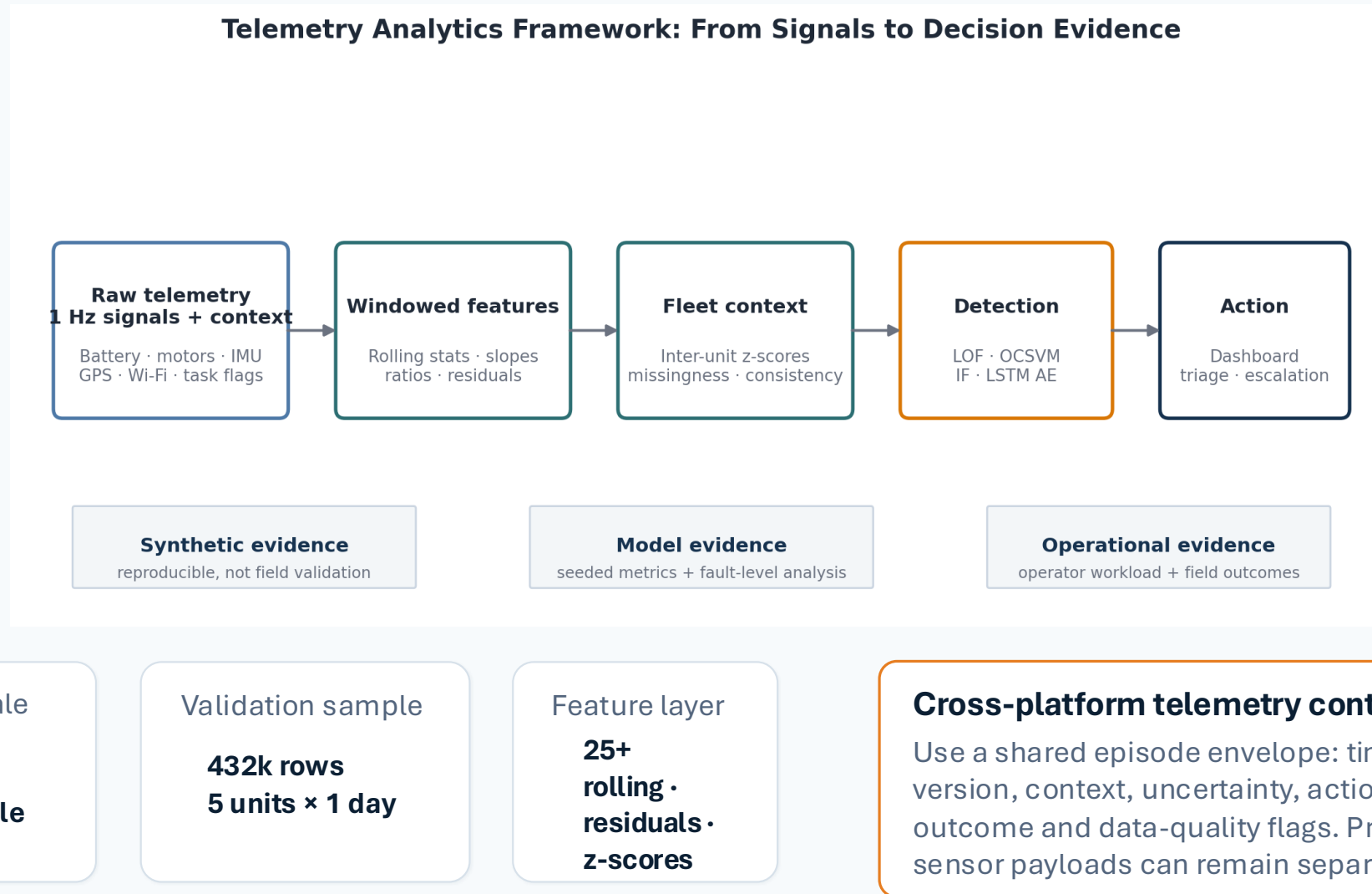
- 1 Do not overcentralize the model**
PIC 2.0 should be an evidence and governance layer, not a one-model-fits-all claim.
- 2 Keep product policies specialized**
Rover, Sentinel, Fari, Senpai and Humanoid need different operating regimes and validation standards.
- 3 Track reuse as a KPI**
Every product release should state what was reused unchanged, adapted, or built product-specifically.

Portfolio Compounding: “One Intelligence Layer, Many Embodiments”

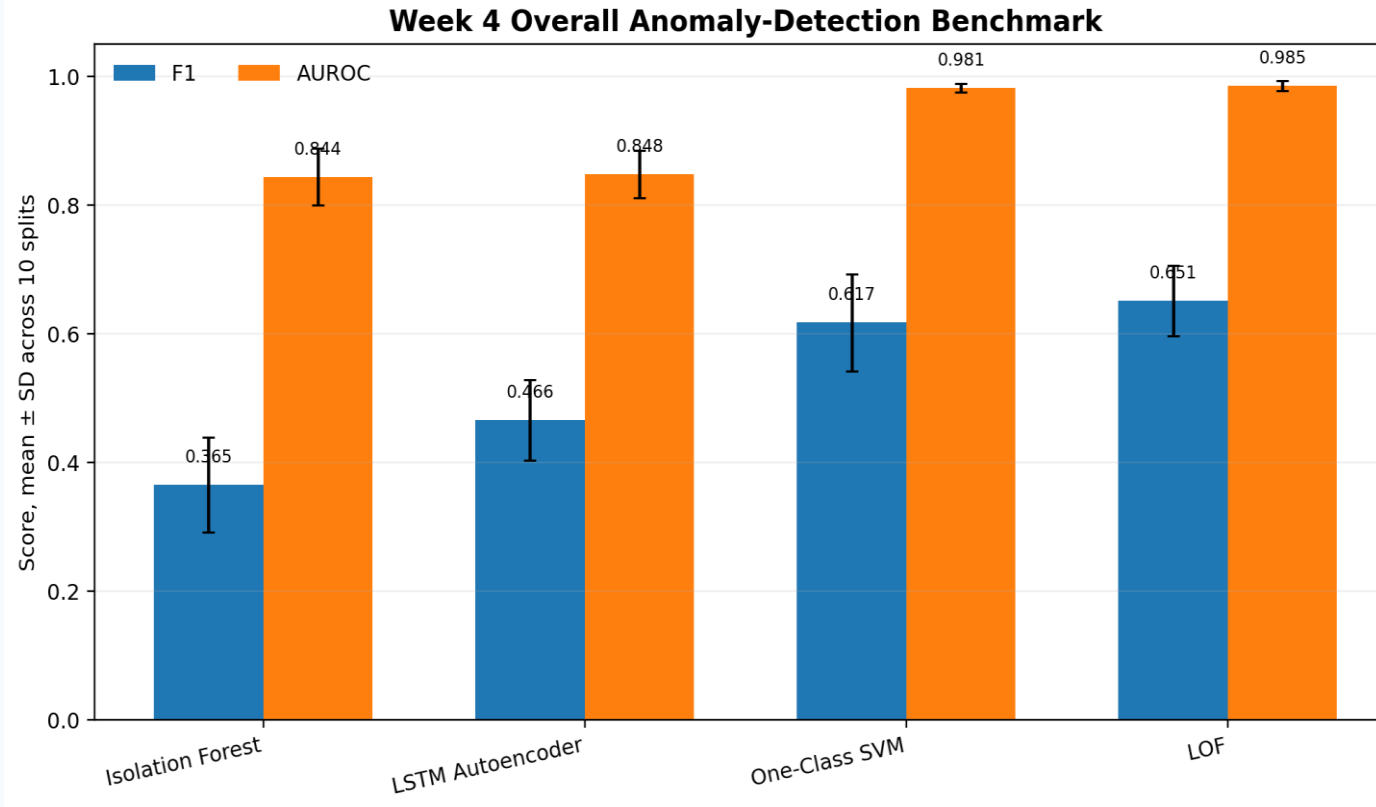


Strategic thesis: reuse telemetry, evaluation contracts, safety rules and deployment evidence across products

Telemetry analytics framework: from raw signals to decision evidence



Anomaly detection: LOF is the default accuracy model



LOF AUROC

0.985
 ± 0.008

LOF recall

0.970
 ± 0.033

LOF F1

0.651
 ± 0.055

OCSVM

close
challenger

Recommended operating model

Accuracy-maximizing: LOF.

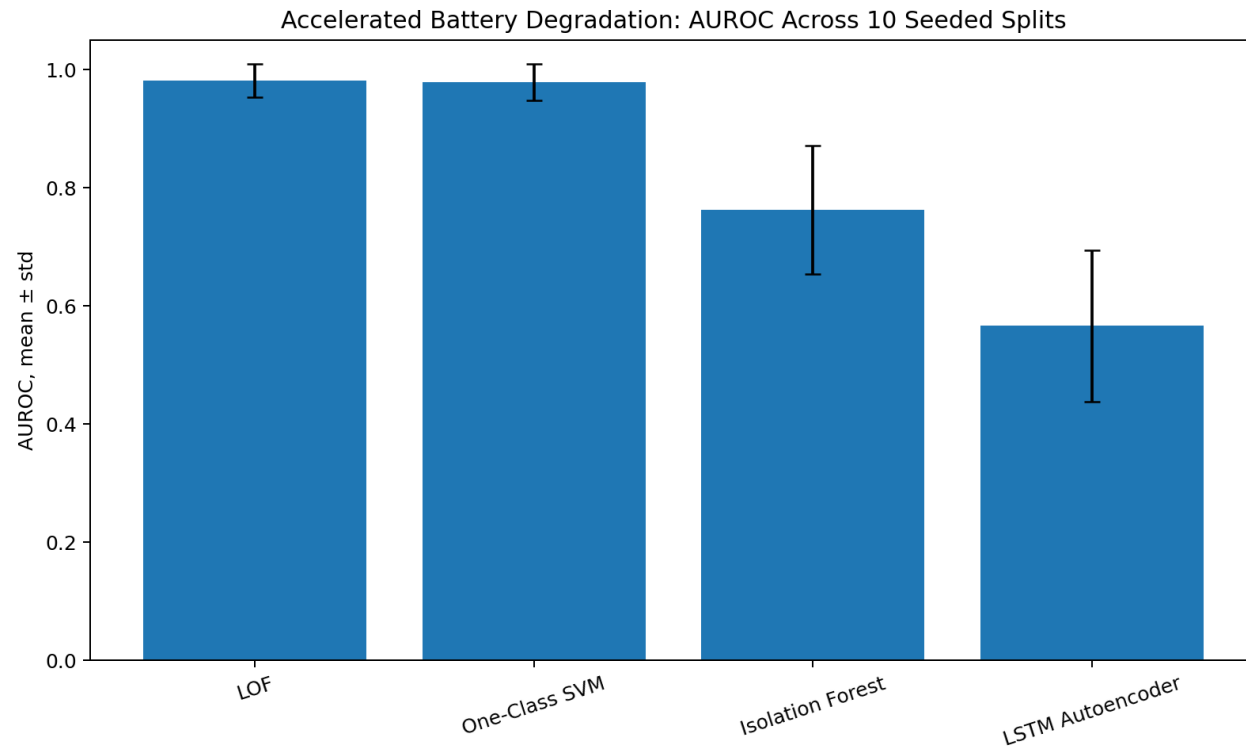
Latency-constrained: Isolation Forest only as first-pass screening.

High-stakes escalation: statistical detector + deterministic guards + human review.

Why it matters

Benchmark evidence supports detector choice by regime, not a single “winning algorithm” across every operational need.

Supervisor follow-up: LSTM AE is not a battery-degradation detector yet



LSTM battery
AUROC

0.566
± 0.128

Events detected

1 / 30
at original
threshold

Zero-recall splits

9 / 10
thresholded
result

Battery-focused
AUROC

0.670
post-hoc score

Interpretation

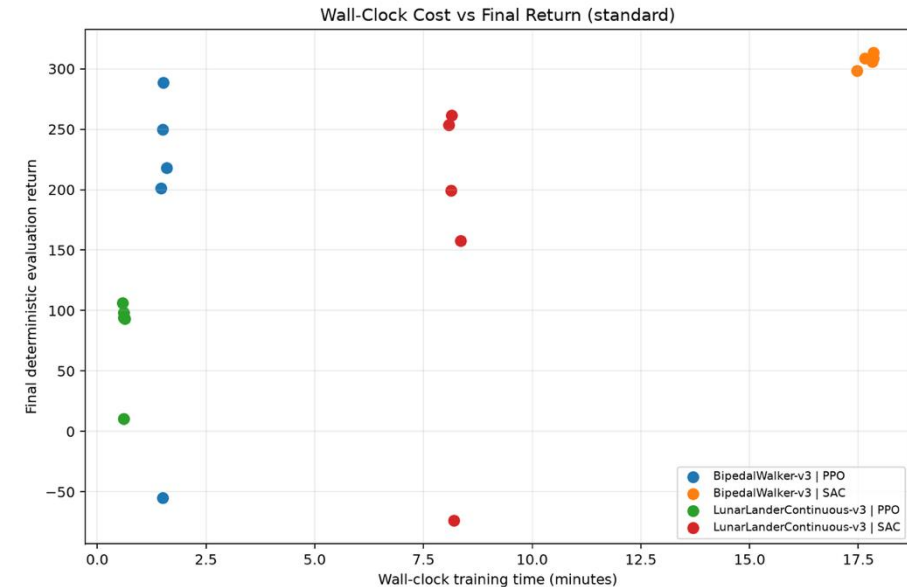
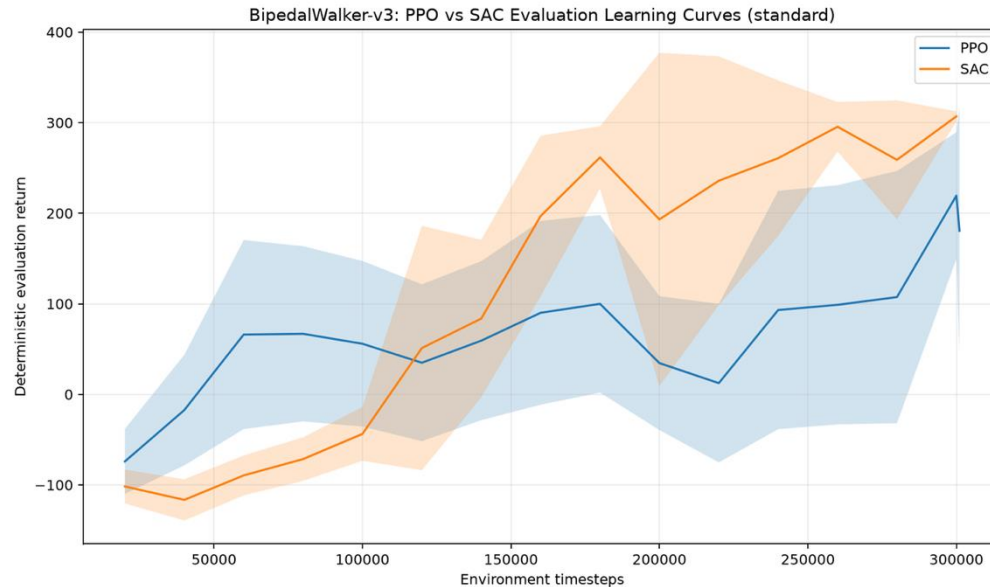
The LSTM contains some battery signal, but equal-weight reconstruction across 17 features dilutes the channel-specific anomaly. Lower thresholds and extra epochs do not resolve the failure.

Capstone impact

Do not market the current LSTM AE as a specialized battery detector. Future work should use charge-cycle horizons, battery-weighted scoring and a dedicated battery-health baseline.

RL benchmarks: learning quality must be tied to operating constraints

Week 5 RL Benchmark: Learning Quality vs. Compute Cost



1

Match method to hierarchy level

High-level planning and low-level control need different objectives, cadence and validation contracts.

2

Do not trust reward alone

Track task success, collisions, regression, runtime, interventions and worst-case behavior.

3

Shared policy is not coordination

Parameter sharing improved distance but increased collisions and did not reach the coordination threshold.

BipedalWalker SAC

307 ± 5.5
final return

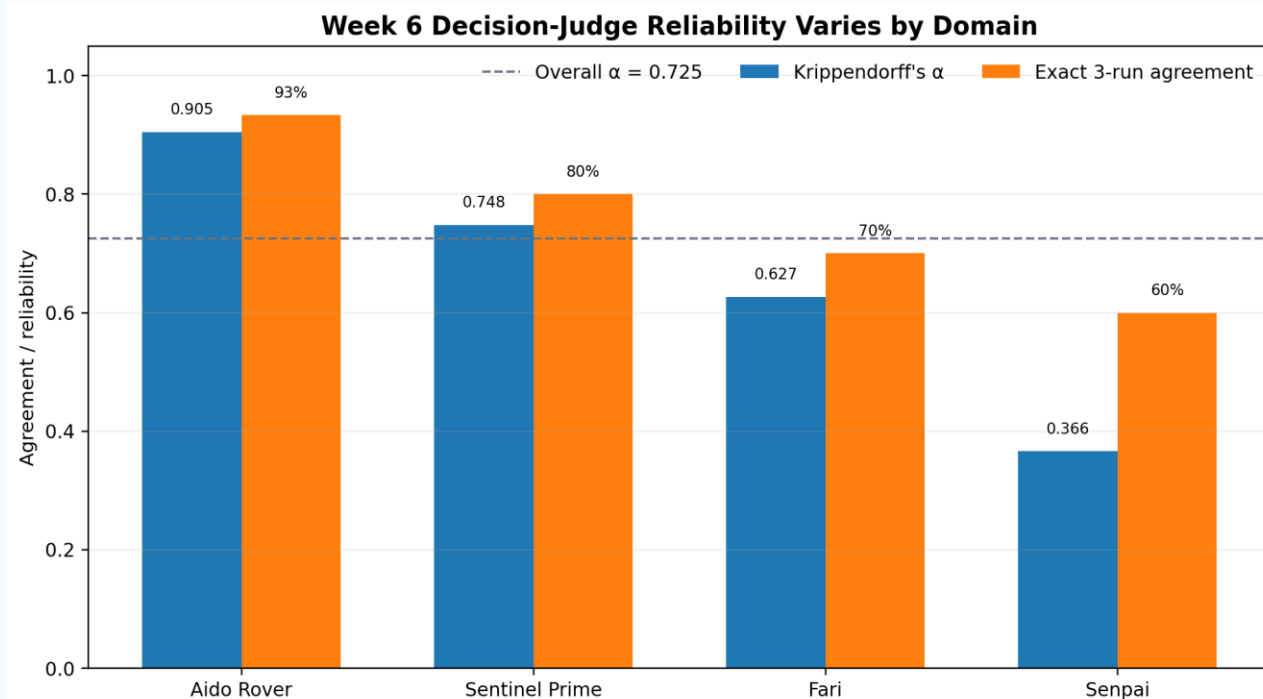
BipedalWalker PPO

91s
fastest runtime

Evidence boundary

These are toy-environment learning benchmarks. They inform HTD-IRL and CRL-MRS design, but they do not prove field-robot readiness.

Decision evaluation: reliability varies sharply by domain



Scenarios

50
YAML bank

Judgments

150
3 judge seedsOverall α **0.725**
tentative

3-run agreement

78%
overall

Rule accuracy

86%
exact match

PASS recall

59.7%
exact matches

Key interpretation

Exact action labels remain authoritative. The LLM judge is best treated as a secondary review signal because agreement is not the same as correctness.

Domain	α	3-run agreement	Use
Aido Rover	0.905	93%	structured operational decisions
Sentinel	0.748	80%	tentative alert escalation review
Fari	0.627	70%	limited semantic care review
Senpai	0.366	60%	poor for automated use

Dashboard overview: one workspace, three decision personas

Audience

Persona 1 **Persona selector** ⓘ

Engineering Manager ▾

Prioritizes detector quality, seed stability, failure modes, and precision/recall tradeoffs.

Data status

Weeks 3–6 sample data loaded

All fleet telemetry, alerts, and operational examples are synthetic. No real customer or

ANALYST WORKSPACE

InGen Fleet, Policy & Decision Intelligence

One dashboard for fleet-health monitoring, reinforcement-learning policy comparison, and decision-evaluation reliability.

2 **Workspace navigation**

☒ Fleet Health ☐ Policy Benchmark ☐ Decision Evaluation

Recommended starting workspace for Engineering Manager: Fleet Health. You can still open any workspace manually.

3 **Persona-specific lens**

Engineering Manager lens · Fleet Health

Prioritizes detector quality, seed stability, failure modes, and precision/recall tradeoffs.

4 **At-a-glance KPI strip**

Fleet units

5

Units needing attention

3

Best anomaly F1

0.651

↑ LOF

Best anomaly AUROC

0.985

Sample files

23 public-safe

Fleet units

5 snapshot

Decision scenarios

50 AMDC bank

Seeded judgments

150 FLAN-T5

Design principle

Persona-aware, not role-restrictive. The dashboard changes the recommended starting workspace and emphasis while preserving access to all evidence.

Persona	Starts with	Primary decision
Engineering	Fleet Health	Which detector is accurate and stable?
Product	Policy Benchmark	What performs best at what cost?
Customer Success	Fleet Health	Which unit needs attention and what next?

Evidence labeling protects trust

The dashboard separates synthetic fleet state, benchmark outputs and reliability metrics so a polished UI does not become an accidental production claim.

My platform recommendations: reuse the evidence spine, specialize the products

Platform	My recommended next priority	Suggested evidence
PIC 2.0	Shared telemetry + evaluation registry	Reuse ratio, regression pass rate, rollback readiness
Aido Rover	Fleet-health service + energy-aware constraints	Alert precision/recall, intervention, uptime, energy reserve
Sentinel Prime AI	Operator trust over raw detector sensitivity	False alerts/hour, ECE, review time, override rate
Fari	Safe autonomy boundary + caregiver escalation	Engagement, handoff success, unnecessary escalation, consent violations
Senpai	Learning outcomes + safeguarding regression suite	Learning gain, misconception correction, intervention, safeguard pass rate
Aido Humanoid	Skill-based evidence ladder + sim-to-real discipline	Task success, interventions, collisions, transfer gap, runtime

My strongest recommendation

Invest first in the shared evidence infrastructure before expanding the model portfolio in parallel. This converts breadth from engineering complexity into a compounding platform advantage.

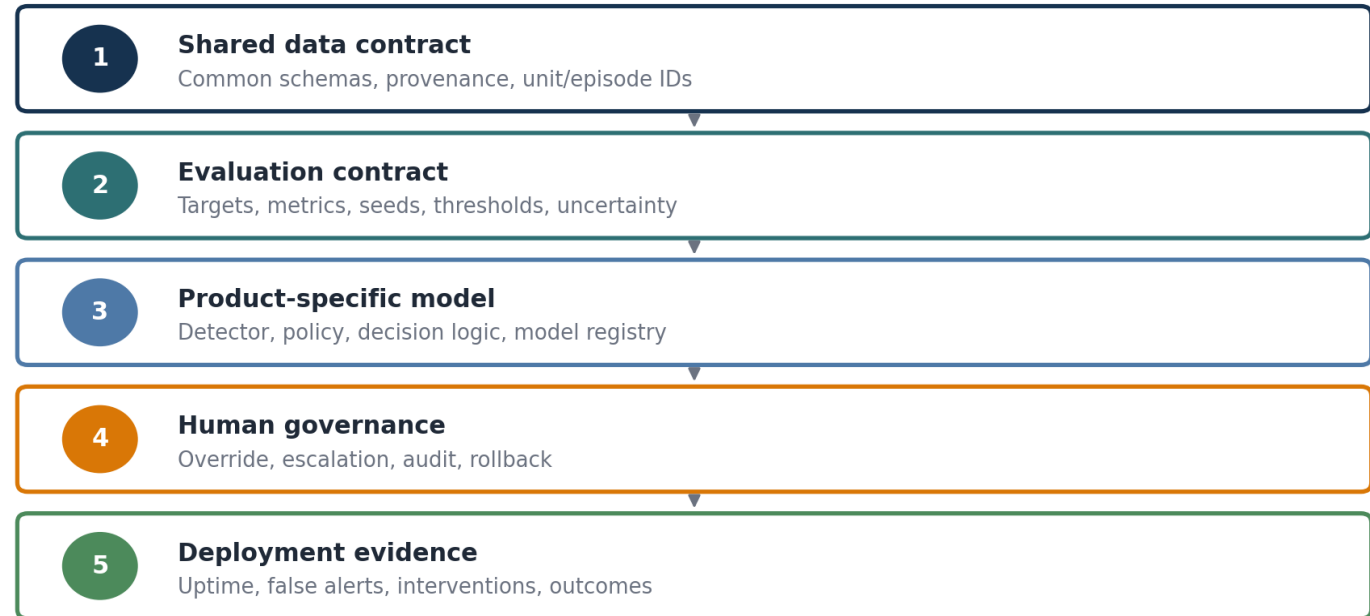
Next steps: build the evidence ladder before scaling autonomy claims

Key limitations

Synthetic telemetry and injected faults; toy RL environments; analyst-defined decision labels; conservative small LLM judge; dashboard is local CSV snapshot, not live operations.

- 1 0–3 months**
Standardize cross-product telemetry and decision episode schema.
- 2 3–6 months**
Operationalize Rover + Sentinel with longer telemetry and operator review.
- 3 6–9 months**
Add human-reviewed Fari/Senpai scenario labels and outcome rubrics.
- 4 9–12 months**
Build Humanoid skill library, sim-to-real stress tests and rollback gates.

Recommended Cross-Portfolio Operating Model



Closing recommendation

One cross-portfolio data/evaluation spine · product-specific models and safeguards · disciplined evidence ladder from synthetic benchmarks to field outcomes.